

A Weyl extremal equality without a common eigenbasis

Counterexample to Conjecture 00000007152

ziangni-sys 4 October 2026

The extremal equality

For real symmetric matrices, the largest-eigenvalue Weyl inequality is

$$\lambda_{\max}(A + B) \leq \lambda_{\max}(A) + \lambda_{\max}(B).$$

We refute the source's necessity of a common eigenbasis for Weyl extremal equality by exhibiting equality in this particular bound without such a basis. The example does not assert simultaneous saturation of every Weyl inequality; that is a stronger condition than equality in this extremal bound.

Consider the actual symmetric matrices and vector

$$A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad e = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Both $Ae = 3e$ and $Be = 3e$, so $(A + B)e = 6e$.

Actual largest eigenvalues

Write $v = (v_0, v_1, v_2)$ and $S(v) = v_0^2 + v_1^2 + v_2^2$. Then

$$v^T Av = 3v_0^2 + v_1^2 \leq 3S(v), \quad v^T Bv = 3v_0^2 + (v_1 + v_2)^2 \leq 3S(v).$$

The second bound follows from $(v_1 + v_2)^2 \leq 2(v_1^2 + v_2^2)$, since $(v_1 - v_2)^2 \geq 0$. Adding the two bounds gives $v^T(A + B)v \leq 6S(v)$. For any nonzero real eigenvector $Mv = \mu v$, the quadratic form is $v^T Mv = \mu S(v)$ and $S(v) > 0$. These bounds therefore imply that every eigenvalue of A , B , and $A + B$ is at most 3, 3, and 6, respectively. Together with the displayed eigenvector e , they prove

$$\lambda_{\max}(A) = 3, \quad \lambda_{\max}(B) = 3, \quad \lambda_{\max}(A + B) = 6 = 3 + 3.$$

This verifies the required equality using the matrices themselves, rather than assuming an eigenvalue table.

No common eigenbasis

If a basis $\{b_i\}$ consisted of simultaneous eigenvectors, with $Ab_i = \alpha_i b_i$ and $Bb_i = \beta_i b_i$, then

$$ABb_i = \beta_i \alpha_i b_i = \alpha_i \beta_i b_i = BAb_i.$$

Linearity and the spanning property of the basis would give $AB = BA$. However, for $w = (0, 0, 1)^T$ direct computation gives

$$ABw = (0, 1, 0)^T, \quad BAw = (0, 0, 0)^T.$$

Thus no common eigenbasis exists, even without requiring the basis to be orthogonal. The matrices share the one extremal eigenvector needed to saturate the top Weyl bound, while their actions on its complementary plane do not commute. A common eigenbasis is consequently not necessary for this Weyl extremal equality.

Formal verification

Lean defines actual matrix multiplication, eigenvalue witnesses, and universal largest-eigenvalue bounds. It checks symmetry and all three largest eigenvalues. The theorem `no_common_eigenbasis` quantifies over an actual Lean `Basis` with arbitrary index type and proves the commutation contradiction. The final theorem packages the exact extremal equality together with the absence of a common eigenbasis.